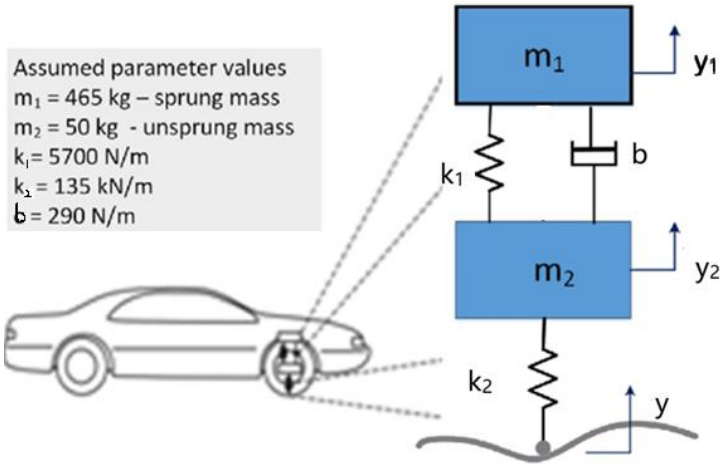
Lab 1

For a 1/4 car model as shown in the figure below, run simulations in Matlab and complete the following tasks:

1) With the transfer functions derived in the class, create the transfer functions in Matlab and simulate the output of the system for the following inputs:
   $y(t)$: step input
   $y(t)$: sinusoidal input
2) With the provided GUI files ("busgui.fig", "busgui.m"), substitute the derived transfer functions into the specified place and generate the car response animations. Adjust the parameters and see the change.




```
73      %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
74      %%%%%%%%%%%% Define the transfer function here %%%%
75 —      m1=465;
76 —      m2=50;
77 —      k1 = 5700;
78 —      k2 = 135000;
79 —      b1 = 290;
80
81 —      numa = [39150000 769500000];
82 —      dena = [23250 149350 65710500 39150000 769500000];
83
84      %%%%%%%%%%%%Do Not change anything below%%%%%%%%%%%%%
85      %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```